Mathematics

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Chapter 1

Groups

1.1 Magmas and monoids

Definition 1.1. A magma is a pair $(S, *)$ with S a set and

$$*: S \times S \to S$$

an internal composition law.

Definition 1.2. A magma $(S, *)$ is said to be associative if $\forall a, b, c \in S,\ (a * b) * c = a * (b * c)$.

Definition 1.3. A magma $(S, *)$ is said to be commutative if $\forall a, b \in S,\ a * b = b * a$.

Definition 1.4. An identity element of a given magma $(S, *)$ is an element $e \in S$, such that $\forall a \in S,\ e * a = a * e = a$.

Theorem 1.5. *If a magma $(S, *)$ has an identity element, then it is unique.*

Proof. If $e, e' \in S$ are identity elements for $*$, then $e = e * e' = e'$. □

Definition 1.6. Given a magma $(S, *)$ with an identity element $e \in S$. An element $a \in S$ is said to be invertible if there exists b such that $a * b = b * a = e$. b is then called an inverse of a.

Definition 1.7. A monoid is an associative magma with an identity element.

Theorem 1.8. *If $(S, *)$ is a monoid, then every invertible element $a \in S$ has a unique inverse, denoted by a^{-1}.*

Proof. Let $b, b' \in S$ be two inverses of a. Then we have

$$b = b * (a * b') = (b * a) * b' = b'.$$

□

Notation 1.9. If $(S, *)$ is a monoid, then for all $a \in S$, we denote:

$$a^0 = e \quad \text{and} \quad a^n = \underbrace{a * \cdots * a}_{n \text{ times}} \quad (\text{for } n \geq 1).$$

Proposition 1.10. *Given a monoid $(S, *)$, and $a, x, y \in S$,*

(i) If $a * x = a * y$ and a is invertible, then $x = y$,

(ii) If x is invertible, then x^{-1} is invertible and $(x^{-1})^{-1} = x$.

(iii) If x and y are invertible, then $x * y$ is also invertible and $(x * y)^{-1} = y^{-1} * x^{-1}$,

(iv) If x is invertible, then for all $n \in \mathbb{N}$, x^n is also invertible and $(x^n)^{-1} = (x^{-1})^n$. Hence we can denote $x^{-n} = (x^n)^{-1}$,

Proof. (i) $x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y$,

(ii) Direct consequence of the invertible element definition,

(iii) $(x * y) * (y^{-1} * x^{-1}) = x * y * y^{-1} * x^{-1} = e$ and similarly for the other way,

(iv) It suffices to combine the two preceding properties in a proof by induction.

□

1.2 Groups

Definition 1.11. A group is a monoid such that every element is invertible.